

VaR vs Expected Shortfall:

Coherence, Regulation, and Applications in Mortgage Credit Risk

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April 27, 2026

Abstract

Regulatory frameworks for financial risk have migrated from Value-at-Risk (VaR) to Expected Shortfall (ES) in market risk, with the Basel Committee's Fundamental Review of the Trading Book completing the transition in January 2019. Insurance frameworks are split: Solvency II retains VaR at 99.5%, while the Swiss Solvency Test uses ES at 99%. This paper develops the technical case behind the shift. It presents the four coherence axioms of Artzner et al. (1999), shows with a minimal two-asset counterexample that VaR fails subadditivity, and reviews Acerbi and Tasche (2002) on the coherence of ES. It derives closed-form expressions for ES under the Asymptotic Single Risk Factor (ASRF) model that underlies Basel II's IRB formulas, and validates the derivation on a synthetic mortgage portfolio of 100,000 loans. The ES/VaR ratio is shown to decrease with the confidence level α under Gaussian factor dynamics, a model-specific feature rather than a property of credit portfolios, as the sequel of this series will establish. Applications in credit portfolio measurement are illustrated under three regulatory measures (VaR 99.5%, ES 99%, ES 97.5%) on the same portfolio. The paper is positioned as a technical exposition: it documents properties, it does not prescribe practice.

1 Introduction

A risk manager confronting a loss distribution asks two questions. The first is *what is the worst plausible outcome?*, a quantile question, naturally answered by Value-at-Risk. The second is *if things do go badly, how badly?*, an expectation question, naturally answered by Expected Shortfall. Banking and insurance regulation has so far privileged the first question: Basel II's Internal Ratings-Based capital charge, Solvency II's Standard Formula for the Solvency Capital Requirement, and the Mexican framework for maximum retention limits all anchor to a quantile of the loss distribution.

The limitations of the first question have been understood since the late 1990s. Artzner et al. (1999) showed that VaR fails subadditivity (the property that combining two portfolios should not increase the total risk). They introduced the axiomatic framework of *coherent risk measures*. Acerbi and Tasche (2002) proved that ES satisfies all four coherence axioms. The market-risk community absorbed these findings slowly at first and then all at once: the 2007–2009 crisis exposed failures in VaR-based trading book capital, the Basel Committee responded with the Fundamental Review of the Trading Book (Basel Committee on Banking Supervision, 2013), and the final rules (Basel Committee on Banking Supervision, 2019) made ES at 97.5% the regulatory standard for market risk. Insurance regulation has not converged: Solvency II kept VaR at 99.5%, while the Swiss Solvency Test adopted ES at 99% over a decade earlier.

This paper develops the technical case behind the shift, without prescribing practice for any particular insurance framework. Section 2 fixes notation. Section 3 presents the coherence axioms and a minimal counterexample showing VaR's failure of subadditivity. Section 4 derives ES under the ASRF model in closed form. Section 5 validates the derivation on a synthetic

mortgage portfolio. Section 6 reviews the regulatory migration from VaR to ES in market risk. Section 7 applies the machinery to two questions in mortgage credit insurance, retention sizing and solvency capital, under three regulatory measures. Section 8 notes that the quantitative results are conditional on the Gaussian factor assumption, which is relaxed in the sequel.

The paper is the third in a series. Post 1 derived the IRB formula from first principles. Post 2 developed Monte Carlo techniques for portfolio loss simulation, including importance sampling and quasi-Monte Carlo.

2 Value-at-Risk and Expected Shortfall

2.1 Definitions

Let L denote the loss random variable of a portfolio over a fixed horizon, with loss distribution function $F_L(\ell) = \mathbb{P}(L \leq \ell)$.

Definition 1 (Value-at-Risk). For a confidence level $\alpha \in (0, 1)$, the Value-at-Risk of L at level α is

$$\text{VaR}_\alpha(L) = \inf\{\ell \in \mathbb{R} : F_L(\ell) \geq \alpha\} = F_L^{-1}(\alpha). \quad (1)$$

When F_L is continuous and strictly increasing, this coincides with the α -quantile.

Definition 2 (Expected Shortfall). The Expected Shortfall of L at level α is

$$\text{ES}_\alpha(L) = \frac{1}{1 - \alpha} \int_\alpha^1 \text{VaR}_u(L) du. \quad (2)$$

Definition 2 is the definition of Acerbi and Tasche (2002), robust to distributions with atoms. When F_L is continuous at VaR_α , the integral representation coincides with the more intuitive *tail conditional expectation*

$$\text{ES}_\alpha(L) = \mathbb{E}[L \mid L \geq \text{VaR}_\alpha(L)]. \quad (3)$$

For discrete or mixed distributions the two quantities differ, and only the integral form in (2) is coherent in general.

2.2 Intuitive reading

VaR answers a question about a single quantile of the loss distribution: the *edge* of the tail at level α . ES answers a question about the tail itself: the average loss conditional on being in the worst $(1 - \alpha)$ fraction of outcomes. The two questions sound similar and in many practical settings deliver similar numbers, but they describe genuinely different quantities. VaR locates a threshold; ES summarises what happens past the threshold.

The distinction matters whenever the tail has structure. Two portfolios with identical 99th-percentile losses can have very different conditional mean losses in the tail beyond that point: one with losses clustered just past the threshold, the other with a long heavy tail extending far beyond. VaR scores both portfolios as equally risky; ES scores them differently. For a risk manager deciding how much capital to set aside, the second answer is the actionable one, because actual losses in adverse scenarios will be drawn from the entire tail, not from the threshold itself.

The same argument applies to the comparison of risk measures across portfolios. A portfolio whose loss distribution has a thin tail and a portfolio whose loss distribution has a fat tail

can produce identical VaR figures at any single chosen confidence level, yet the second carries materially more risk in any operationally meaningful sense. The discrepancy is not a defect of the quantile — the quantile reports correctly what it is asked to report — but a feature of summarising a distribution by one of its points rather than by an expectation. ES integrates the tail; VaR samples it.

A simple numerical contrast makes the gap visible. Consider a portfolio whose loss at the 99% confidence level is one billion currency units. If the distribution beyond that threshold is thin — losses concentrated just past one billion, falling rapidly thereafter — the conditional mean of losses in the tail might be 1.05 or 1.1 billion, only marginally above the threshold. If the distribution is heavy-tailed, the same 99% quantile may sit alongside a tail mean of 1.5 or 2 billion: half of all adverse-scenario losses lie substantially above the quantile, and ES captures the average of those losses while VaR remains pinned to the threshold. A capital allocator working from VaR alone would treat both portfolios identically; an allocator working from ES would set aside meaningfully more for the second. This is not a corner case: heavy-tailed loss distributions are the rule rather than the exception in credit and operational risk, where defaults and operational failures cluster during stress periods rather than spreading uniformly across time.

This is not merely a pedagogical observation. As Section 3 will show, the failure of VaR to summarise the tail is closely related to its failure to satisfy a fundamental axiom of risk-measure consistency, and the practical consequences of that failure motivated the regulatory migration documented in Section 6. Before turning to either the coherence argument or the regulatory record, we address the property that makes the comparison between VaR and ES tractable in the first place: under the asset-value framework that underlies modern credit risk capital formulas, both measures admit closed-form expressions, and their numerical behaviour can be characterised exactly, as Section 5 illustrates on a 100,000-loan synthetic portfolio that we revisit throughout the paper.

3 Coherence and the Failure of Subadditivity

Artzner et al. (1999) proposed an axiomatic framework that characterises risk measures consistent with basic economic intuition. A risk measure is a mapping $\rho : \mathcal{X} \rightarrow \mathbb{R}$ that assigns a real-valued capital requirement to each risk position X in a space $\mathcal{X}$ of random variables. The signs in what follows treat L as loss, so $\rho(L) \geq 0$ represents required capital.

3.1 The coherence axioms

Axiom 1 (Translation invariance). For any constant $c \in \mathbb{R}$, $\rho(L + c) = \rho(L) + c$. Adding a deterministic loss increases required capital by the same amount.

Axiom 2 (Monotonicity). If $L_1 \leq L_2$ almost surely, then $\rho(L_1) \leq \rho(L_2)$. A pointwise larger loss should not carry a smaller capital charge.

Axiom 3 (Positive homogeneity). For any $\lambda \geq 0$, $\rho(\lambda L) = \lambda \rho(L)$. Scaling the exposure scales the capital charge proportionally.

Axiom 4 (Subadditivity). For any two loss random variables L_1, L_2 , $\rho(L_1 + L_2) \leq \rho(L_1) + \rho(L_2)$. Combining two portfolios cannot increase total required capital beyond the sum of the individual requirements.

A risk measure satisfying all four axioms is called *coherent*. The first three are uncontroversial.

Subadditivity is the mathematical formalisation of the diversification principle: merging exposures should, at worst, leave total capital unchanged, never raise it. A risk measure that violates subadditivity can produce the perverse recommendation that splitting a portfolio into two pieces reduces required capital. This is a regulatory arbitrage that undermines the economic meaning of the measure. The point is not academic. A capital framework anchored on a non-coherent measure can in principle be gamed by structural decisions (splitting books, creating subsidiaries, or repackaging exposures across legal entities) that do not change the underlying risk but lower reported capital. Coherence rules out that family of arbitrages by construction. The four axioms of Artzner et al. (1999) are the minimal set under which the capital charge of a portfolio cannot be reduced by reorganisation alone — a property that any candidate regulatory measure should satisfy. The next subsection shows that VaR does not, even in the simplest credit-risk setting.

3.2 VaR fails subadditivity: a two-bond example

The failure of VaR as a coherent risk measure is not a curiosity of pathological distributions. It arises in the simplest credit-risk setting: two independent bonds with low default probabilities. The following example appears in McNeil et al. (2015) and is reproduced here for completeness.

Example 1 (Two-bond counterexample). Consider two independent bonds, A and B , each with face value 100, default probability $p = 4\%$, and loss given default equal to the face value. The loss on each bond is

$$L_A, L_B \in \{0, 100\}, \quad \mathbb{P}(L_A = 100) = \mathbb{P}(L_B = 100) = 0.04,$$

with L_A independent of L_B .

VaR at 95%. Since $\mathbb{P}(L_A = 0) = 0.96 \geq 0.95$, the 95%-quantile of each bond's loss distribution is zero:

$$\text{VaR}_{0.95}(L_A) = \text{VaR}_{0.95}(L_B) = 0.$$

Now consider the combined portfolio $L_A + L_B \in \{0, 100, 200\}$ with probabilities

$$\mathbb{P}(L_A + L_B = 0) = 0.9216, \quad \mathbb{P}(L_A + L_B = 100) = 0.0768, \quad \mathbb{P}(L_A + L_B = 200) = 0.0016.$$

The cumulative probability at zero is $0.9216 < 0.95$, so

$$\text{VaR}_{0.95}(L_A + L_B) = 100.$$

Failure of subadditivity. The combined VaR exceeds the sum of the individual VaRs by the full face value of one bond:

$$\text{VaR}_{0.95}(L_A + L_B) = 100 > 0 = \text{VaR}_{0.95}(L_A) + \text{VaR}_{0.95}(L_B).$$

A capital allocator using VaR at 95% concludes that merging the two bonds into a single book increases capital requirements, even though the portfolio is objectively more diversified than either component alone.

ES preserves subadditivity. Using Definition 2, one can verify that

$$\text{ES}_{0.95}(L_A) = \text{ES}_{0.95}(L_B) = 80, \quad \text{ES}_{0.95}(L_A + L_B) = 160,$$

so the subadditivity inequality holds with equality. More generally, for any two integrable losses and any $\alpha \in (0, 1)$, $\text{ES}_\alpha(L_1 + L_2) \leq \text{ES}_\alpha(L_1) + \text{ES}_\alpha(L_2)$.

The mechanism behind VaR's failure is discrete tail concentration: when each bond's default event sits just below the threshold, the rare event that either bond defaults crosses the threshold in the aggregate, and VaR registers the jump while individual VaRs do not. The same mechanism recurs in real portfolios. The probabilities are not identical in practice, but whenever tail events concentrate near the quantile.

3.3 Elicitability and what it means for backtesting

Coherence is not the only property that matters in practice. A risk measure also has to be *verifiable after the fact*: once losses are observed, one needs a statistical procedure to ask whether the forecast made a year ago was reasonable. The technical name for the property that makes such verification clean is *elicibility*.

The idea is simple in everyday terms. Suppose a forecaster reports a single number r as a risk prediction, and we want to grade that report after the loss is realised. We need a *scoring rule* $S(r, L)$, a function of the forecast r and the realised loss L , that assigns a penalty whenever the forecast is wrong. Now imagine we could replay history many times with the same underlying loss distribution; the forecaster's *expected* score under that distribution depends on what value r they reported. The rule is called *strictly consistent* for a risk measure ρ if, holding the true loss distribution fixed, the expected score is minimised only when the forecaster reports the correct value $r = \rho(L)$, and is strictly higher for any other report. Put differently: under a strictly consistent rule, the only way a forecaster can minimise their expected penalty is to report the true risk measure. They cannot game the rule by reporting a different number.

A concrete example: if we grade a forecast of the 95% quantile with the weighted absolute-error rule

$$S(r, L) = (\mathbf{1}_{\{L \leq r\}} - 0.95) \cdot (r - L),$$

a direct calculation shows that the forecaster's expected score is minimised exactly at $r = \text{VaR}_{0.95}(L)$. Reporting any other number gives a strictly higher expected penalty. This is the concrete sense in which quantiles admit an objective grading rule.

A risk measure admitting such a rule is called *elicitable*. The practical consequence is that elicibility gives us an objective way to rank two risk forecasts based purely on observed outcomes, without needing to know the true loss distribution: whoever reports lower average scores over a long enough sample is the better forecaster.

Gneiting (2011) proved that quantiles, and therefore VaR, are elicitable. The consistent scoring rule is a weighted asymmetric absolute error: under-predicting a loss at confidence level α is penalised more heavily than over-predicting, with weights chosen so that the optimal report is exactly the α -quantile. Gneiting also proved the negative result that ES is *not* elicitable: no scoring rule of the form $S(r, L)$ exists whose expected value is minimised at $r = \text{ES}_\alpha(L)$. For a few years this led some authors to argue that regulators should retain VaR on backtesting grounds alone.

The debate was settled in two steps. First, Emmer et al. (2015) argued that elicibility matters for comparing competing forecasts (which model is more accurate?), but it is not decisive for the choice of a regulatory risk measure (which quantity should capital reflect?). The two questions are distinct: a non-elicitable measure can still be backtested, just not through a single score

function. Second, they showed that the *pair* $(\text{VaR}_\alpha, \text{ES}_\alpha)$ is jointly elicitable: consistent two-dimensional scoring rules exist that deliver both numbers at once. This opens the door to rigorous backtesting procedures for ES that piggy-back on the quantile forecast.

This is exactly what the Basel Committee adopted. Rather than backtest ES directly, the FRTB framework (Basel Committee on Banking Supervision, 2019) monitors the underlying VaR forecast at the trading-desk level, counting exceedences against the 97.5% and 99% quantiles. Desks whose VaR model passes the test are allowed to use their internal ES computation for capital; desks whose model fails receive a regulatory add-on to their ES. The elicibility objection was not ignored, it was circumvented: VaR is kept as the backtestable scaffold, and ES is the capital measure built on top of it.

4 Expected Shortfall under the ASRF Model

4.1 The ASRF setup

The Asymptotic Single Risk Factor model, introduced by Vasicek (2002) and formalised by Gordy (2003), underlies the Basel II Internal Ratings-Based approach. Each borrower $i = 1, \dots, N$ is characterised by a probability of default PD_i , a loss given default LGD_i , and an exposure at default EAD_i . Asset returns follow a one-factor structure

$$A_i = w_i Z + \sqrt{1 - w_i^2} \varepsilon_i, \quad Z, \varepsilon_1, \dots, \varepsilon_N \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad (4)$$

where Z is the systematic factor common to all borrowers, ε_i is an idiosyncratic shock, and $w_i = \sqrt{\rho_i}$ is the factor loading with ρ_i the asset correlation. Default occurs when A_i falls below a threshold $d_i = \Phi^{-1}(\text{PD}_i)$, calibrated so that unconditional default probability equals PD_i .

Conditional on the systematic realisation $Z = z$, the default probability of borrower i is

$$p_i(z) = \mathbb{P}(A_i < d_i \mid Z = z) = \Phi\left(\frac{d_i - w_i z}{\sqrt{1 - w_i^2}}\right), \quad (5)$$

and conditional defaults are independent across i .

The portfolio loss is $L = \sum_{i=1}^N \ell_i \mathbf{1}\{A_i < d_i\}$ with $\ell_i = \text{LGD}_i \cdot \text{EAD}_i$. Conditional on $Z = z$, the expected loss is

$$\mu_N(z) = \mathbb{E}[L \mid Z = z] = \sum_{i=1}^N \ell_i p_i(z). \quad (6)$$

Gordy (2003) showed that as $N \rightarrow \infty$ under appropriate name-concentration conditions, $L/N \rightarrow \mu_N(Z)/N$ almost surely. The idiosyncratic component is diversified away and only the systematic factor remains relevant. For finite but large N (such as $N = 10^5$ here), the approximation $L \approx \mu_N(Z)$ is accurate to within Monte Carlo error, a point we verify numerically in Section 5.

4.2 VaR in closed form

Because $p_i(z)$ is a decreasing function of z (a worse economic environment, lower z , raises every borrower's conditional default probability), the conditional loss function $\mu_N(z)$ is strictly decreasing in z . This monotonicity is what makes the IRB formula collapse to a single evaluation. Since μ_N is monotone and $Z \sim \mathcal{N}(0, 1)$, exceeding the α -quantile of the (asymptotic) loss is

equivalent to Z falling below the corresponding Gaussian quantile:

$$\mathbb{P}(L > \ell) = \mathbb{P}(Z < \mu_N^{-1}(\ell)) = \Phi(\mu_N^{-1}(\ell)). \quad (7)$$

Setting the right-hand side equal to $1 - \alpha$ and inverting:

$$\text{VaR}_\alpha(L) = \mu_N(z_\alpha), \quad z_\alpha = \Phi^{-1}(1 - \alpha). \quad (8)$$

Equation (8) is written in the compact form that makes the derivation transparent, but it is perhaps more familiar in the expanded form that appears in the Basel II text. Recall that the conditional default probability of borrower i is $p_i(z) = \Phi\left((d_i - w_i z)/\sqrt{1 - w_i^2}\right)$ with threshold $d_i = \Phi^{-1}(\text{PD}_i)$ and factor loading $w_i = \sqrt{\rho_i}$. Evaluating at the Gaussian quantile $z_\alpha = \Phi^{-1}(1 - \alpha) = -\Phi^{-1}(\alpha)$, and summing over borrowers with $\mu_N(z_\alpha) = \sum_i \text{LGD}_i \cdot \text{EAD}_i \cdot p_i(z_\alpha)$, one recovers the conventional IRB formula

$$K_i = \text{LGD}_i \cdot \Phi\left(\frac{\Phi^{-1}(\text{PD}_i) + \sqrt{\rho_i} \Phi^{-1}(\alpha)}{\sqrt{1 - \rho_i}}\right) - \text{LGD}_i \cdot \text{PD}_i, \quad (9)$$

where K_i is the unexpected-loss capital charge per unit of exposure at default for borrower i . Two things to notice. First, the sign flip: the Gaussian quantile $z_\alpha = \Phi^{-1}(1 - \alpha)$ is negative (deep tail of Z), and the identity $-\Phi^{-1}(1 - \alpha) = \Phi^{-1}(\alpha)$ is what converts the lower-tail event on Z into the upper-tail event on losses, giving the familiar plus sign inside the argument of Φ . Second, the subtraction of $\text{LGD}_i \cdot \text{PD}_i$ at the end isolates the *unexpected* component of the loss: the expected portion is already provisioned through accounting reserves, and regulatory capital is meant to cover only the excess. At $\alpha = 0.999$ (the Basel II horizon) this is the formula the first post in the series derived from first principles. The compact form in (8) is the same object; expression (9) just makes the borrower-level decomposition explicit.

4.3 ES in closed form

The integral definition (2) is equivalent, under the monotonicity of μ_N , to an integral over the tail of the systematic factor. Because μ_N is monotone decreasing, exceeding $\text{VaR}_\alpha = \mu_N(z_\alpha)$ is the same event as $Z < z_\alpha$. Hence

$$\text{ES}_\alpha(L) = \frac{1}{1 - \alpha} \mathbb{E}[\mu_N(Z) \mathbf{1}\{Z < z_\alpha\}] = \frac{1}{1 - \alpha} \int_{-\infty}^{z_\alpha} \mu_N(z) \phi(z) dz, \quad (10)$$

where ϕ is the standard Gaussian density.

Expression (10) has a useful reformulation as a weighted sum of bivariate Gaussian probabilities. Substituting (6) and interchanging sum and integral,

$$\begin{aligned} \text{ES}_\alpha(L) &= \frac{1}{1 - \alpha} \sum_{i=1}^N \ell_i \int_{-\infty}^{z_\alpha} \Phi\left(\frac{d_i - w_i z}{\sqrt{1 - w_i^2}}\right) \phi(z) dz \\ &= \frac{1}{1 - \alpha} \sum_{i=1}^N \ell_i \Phi_2(d_i, z_\alpha; \sqrt{\rho_i}), \end{aligned} \quad (11)$$

where $\Phi_2(\cdot, \cdot; r)$ is the bivariate standard Gaussian cumulative distribution function with correlation r . The identity used in the last step is the standard integral representation of the bivariate normal CDF; see McNeil et al. (2015, Appendix A.5).

Representation (11) is theoretically illuminating — each borrower contributes a bivariate tail probability weighted by its loss-given-default exposure — but in practice the direct numerical evaluation of (10) is faster and more transparent. Evaluating $\mu_N(z)$ is a vectorised operation over the N borrowers, costing roughly one millisecond for $N = 10^5$ on modern hardware. Adaptive Gaussian quadrature (`scipy.integrate.quad`) converges to machine precision in under 100 function evaluations, yielding the ES at any single confidence level in about one second.

4.4 Simulation under large- N asymptotics

Exact Monte Carlo simulation of the loss distribution requires generating M scenarios of the systematic factor Z and, for each scenario, N idiosyncratic shocks ε_i followed by the evaluation of the default indicators $\mathbf{1}\{A_i < d_i\}$. With $N = 10^5$ loans and $M = 2 \times 10^4$ scenarios, this approach stores and evaluates 2×10^9 Bernoulli realisations.

The ASRF asymptotic result eliminates the need for the idiosyncratic component. For N large, $L \rightarrow \mu_N(Z)$ almost surely, so a single draw of Z delivers the loss directly through (6):

$$L^{(m)} \approx \mu_N(Z^{(m)}), \quad m = 1, \dots, M. \quad (12)$$

Memory requirements drop from $O(M \cdot N)$ to $O(M)$, and wall-clock time drops by roughly the same factor. The remaining source of error is the deviation of L from $\mu_N(Z)$ for finite N , which by Hoeffding’s inequality decays as $O(N^{-1/2})$ and is negligible at $N = 10^5$.

A note on estimator choice. An earlier companion paper in this series applied importance sampling combined with quasi-Monte Carlo (Halton sequences) to accelerate convergence when simulating at the idiosyncratic level, where rare tail events made plain Monte Carlo inefficient. In the present setting the estimator is already one-dimensional: the LLN has eliminated the idiosyncratic component and only Z requires simulation. More importantly, Monte Carlo is used here for *validation* rather than computation. The closed-form integration via adaptive quadrature already delivers the asymptotic values to machine precision in about one second; the purpose of the simulation is to confirm that both paths converge to the same number. The comparison is cleanest when the simulation estimator is as basic as possible, which is why we employ plain Monte Carlo. Importance sampling over Z and low-discrepancy sequences would further reduce tail-estimator variance and would be appropriate when the target quantity no longer admits a simple closed form and the integrand becomes genuinely harder to evaluate.

The simulation reported above uses the asymptotic form $\mu_N(Z)$ rather than simulating individual Bernoulli defaults, for the validation-focused reasons explained above. In applied work with real portfolios, this is where Monte Carlo matters most. The simulation is typically done loan-by-loan: for each Monte Carlo scenario a factor Z is drawn, each borrower’s conditional default probability is computed from $p_i(Z)$, and defaults are sampled as Bernoulli trials against those probabilities. The resulting loss distribution is atomic by construction, because the portfolio loss is a discrete sum of per-loan losses and takes only finitely many distinct values. This is the practice the two-bond example of the next subsection illustrates in miniature, and it is where the estimator choice discussed there becomes operationally relevant.

The geometry of the closed-form solution, where VaR sits on the curve $\mu_N(z)$ and what region the ES integrates over, is illustrated in Figure 4 of Section 5, alongside the discussion of the decreasing ES/VaR ratio for which it provides the visual intuition.

5 Numerical Results

5.1 Portfolio specification

The synthetic portfolio consists of $N = 10^5$ mortgage loans with parameters drawn from distributions calibrated to reflect a typical Mexican-market mortgage book. Probabilities of default are drawn from a log-normal distribution with median 5% and shape parameter 0.6, clipped to the interval $[0.5\%, 20\%]$. Loss given default is uniform on $[15\%, 55\%]$, matching the wide spread observed in residential mortgage recoveries after accounting for foreclosure costs, legal delays, and property depreciation. Exposure at default is uniform on $[0.3, 4]$ million MXN, reflecting typical mortgage sizes across urban price strata. The asset correlation is set to $\rho = 0.15$, the midpoint of the Basel II mortgage range. Total portfolio characteristics are summarised in Table 1.

Table 1. Synthetic portfolio characteristics.

Number of loans N	100,000
Total exposure at default	215.1 bn MXN
Average probability of default	5.93%
Average loss given default	34.96%
Expected loss	4.46 bn MXN
Asset correlation ρ	0.15

5.2 Closed-form values

Table 2 reports VaR and ES computed via the closed-form expressions (8) and (10). All values are in billions of Mexican pesos. The ES/VaR ratio is reported in the last column.

Table 2. VaR and ES across confidence levels, closed-form values. Losses in bn MXN.

α	VaR	ES	ES/VaR	ES–VaR
0.950	11.49	14.89	1.2959	3.40
0.975	13.87	17.24	1.2432	3.37
0.990	16.99	20.31	1.1954	3.32
0.995	19.33	22.59	1.1691	3.27
0.999	24.62	27.74	1.1266	3.12

The ES/VaR ratio declines monotonically from 1.296 at $\alpha = 0.95$ to 1.127 at $\alpha = 0.999$. This monotone decline is a feature of the Gaussian factor model, not of the portfolio: we return to it in Section 8. The absolute gap between ES and VaR (last column of Table 2) behaves rather differently: it stays close to 3.1–3.4 billion MXN across the entire grid of confidence levels considered here, narrowing only modestly as α rises. The relative gap and the absolute gap therefore tell different stories about the same numbers, and a regulator focused on capital adequacy would naturally look at the latter rather than the former.

Figure 1 puts the comparison in graphical form. It shows where VaR and ES sit on the simulated loss distribution at the 99.9% confidence level: the gap of 3.12 bn MXN between the two is the slice of the tail that VaR ignores and ES integrates. The corresponding plot of the ES/VaR ratio across α is deferred to Section 5.5, where it accompanies the discussion of the ratio’s declining behaviour.

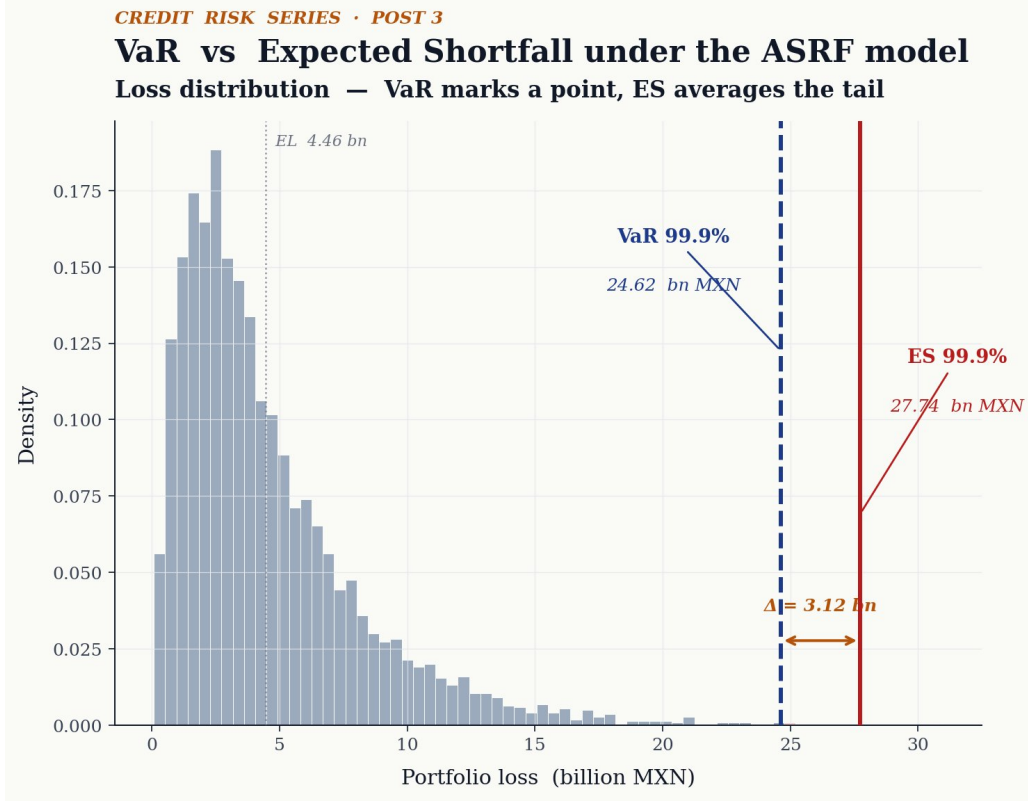


Figure 1. Loss distribution of the 100,000-loan synthetic mortgage portfolio ($\rho = 0.15$, total EAD = 215 bn MXN). VaR at the 99.9% confidence level is marked by the vertical dashed line and ES is reported as the mean of the shaded tail beyond it. The shaded band highlights the entire tail region that ES integrates.

5.3 Monte Carlo validation

To validate the closed-form results, we simulate $M = 20,000$ draws of the systematic factor Z following (12) and take empirical quantiles and tail means of the resulting losses. For each α , the simulation estimator of VaR is the $(\lceil \alpha M \rceil)$ -th order statistic of the simulated losses, and the simulation estimator of ES is the arithmetic mean of the upper $(1 - \alpha)M$ order statistics.

Table 3 compares closed-form and simulation results. All errors fall within $\pm 1\%$ of the analytic values. Figure 2 presents the same comparison graphically.

Table 3. Closed-form vs. Monte Carlo simulation ($M = 20,000$). Losses in bn MXN.

α	VaR			ES		
	Closed form	Simulation	Error (%)	Closed form	Simulation	Error (%)
0.950	11.49	11.49	−0.07	14.89	14.90	0.01
0.975	13.87	13.79	−0.58	17.24	17.28	0.22
0.990	16.99	17.10	0.61	20.31	20.45	0.70
0.995	19.33	19.49	0.85	22.59	22.78	0.83
0.999	24.62	24.54	−0.32	27.74	27.83	0.33

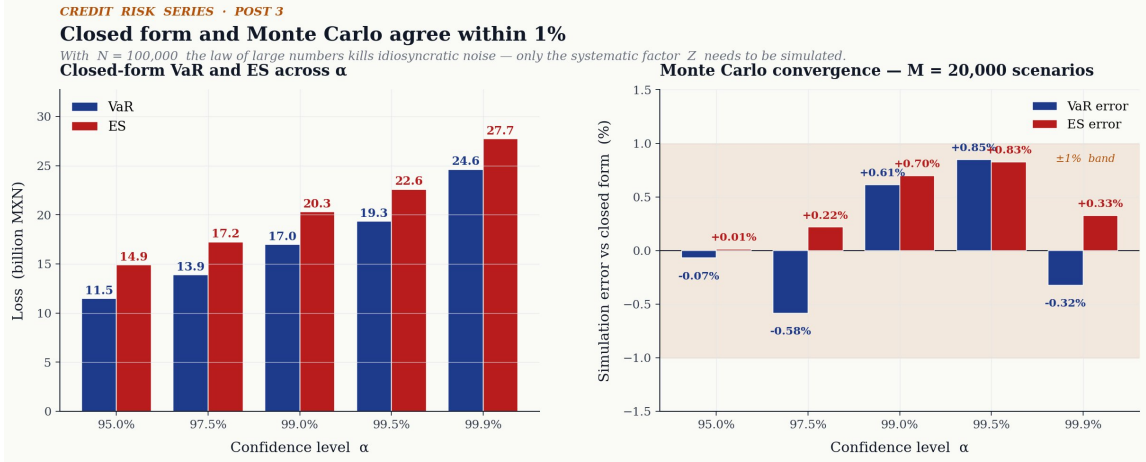


Figure 2. Monte Carlo validation of the closed-form VaR and ES. *Left:* closed-form values at each confidence level. *Right:* simulation error as a percentage of the closed-form value, bounded by $\pm 1\%$ across all tested confidence levels.

5.4 Estimator choice for atomic loss distributions

The Monte Carlo validation of the previous subsection used the simple tail-mean estimator

$$\widehat{\text{ES}}_{\alpha}^{\text{naive}} = \frac{1}{M - \lfloor M\alpha \rfloor} \sum_{i=\lfloor M\alpha \rfloor + 1}^M L_{(i)}, \quad (13)$$

where $L_{(1)} \leq \dots \leq L_{(M)}$ denote the order statistics of M simulated losses. In plain words: sort the simulated losses, drop the first $\lfloor M\alpha \rfloor$ of them, and average what remains. This is the estimator that dominates practice in market-risk historical simulation, and its performance on the ASRF portfolio, reported in Table 3, is satisfactory: errors are bounded by $\pm 1\%$ relative to the closed-form values at $M = 20,000$. There is a reason for this. When F_L is continuous at VaR_{α} , as it is for the asymptotic loss $L = \mu_N(Z)$ (a smooth transformation of a Gaussian factor), the estimator is consistent and well-behaved.

The picture changes when the loss distribution is atomic, which is the generic case whenever one simulates individual Bernoulli defaults rather than invoking the large- N limit. Concretely: if the portfolio has 50 loans and one simulates loss by sampling defaults directly, the resulting loss takes only a finite number of distinct values, and the distribution accumulates probability mass at each of them. Atoms at or near the quantile break the naive estimator in a specific, structural way, and understanding why requires a short detour into how ES is defined when mass sits on the quantile threshold.

The population picture: atoms at the quantile. The two definitions of ES in Section 2, the integral form (2) and the conditional-expectation form (3), coincide when the loss distribution is continuous at VaR_{α} , and Acerbi and Tasche (2002) proved that ES is coherent for any integrable loss under the integral form. When mass sits exactly on the quantile, the two forms diverge, and only the integral form preserves coherence, a distinction that matters precisely when the portfolio is small enough for atoms to be relevant. Suppose F_L has a mass point of size p_0 at $v = \text{VaR}_{\alpha}(L)$, with $F_L(v^-) = \alpha_-$ and $F_L(v) = \alpha_+$, so that $\alpha_- < \alpha \leq \alpha_+$ and $\alpha_+ - \alpha_- = p_0$. The tail that should carry probability $1 - \alpha$ decomposes into the mass strictly above v (probability $1 - \alpha_+$) plus a fraction $\alpha_+ - \alpha$ of the atom itself. The integral form assigns

the atom that fractional weight:

$$\text{ES}_\alpha(L) = \frac{1}{1-\alpha} \left[(\alpha_+ - \alpha) v + \int_{(v, \infty)} \ell dF_L(\ell) \right]. \quad (14)$$

The alternative $\mathbb{E}[L \mid L \geq \text{VaR}_\alpha]$ absorbs the full atom into the tail, inflates the conditioning event to probability $1 - \alpha_-$, and assigns the atom weight p_0 rather than $\alpha_+ - \alpha$. When $p_0 = 0$ the two expressions coincide; when $p_0 > 0$ they differ, and only (14) is the ES of Definition 2. In what follows, “Expected Shortfall” always means the integral form.

The two-bond counterexample of Example 3.2 illustrates the decomposition concretely. The aggregate loss $L_A + L_B$ has a mass point of size $p_0 = 0.0768$ at $v = 100$, with $\alpha_- = 0.9216$ and $\alpha_+ = 0.9984$. At $\alpha = 0.95$ the integral form assigns the atom fractional weight $(\alpha_+ - \alpha)/(1 - \alpha) = 0.0484/0.05 = 96.8\%$ of its normalisation capacity, giving $\text{ES}_{0.95}(L_A + L_B) = 103.2$. The conditional-expectation form assigns the full atom and normalises by $1 - \alpha_- = 0.0784$, yielding 102.04. The numerical gap is modest in this particular configuration, but the structural distinction is general: the fractional weight $(\alpha_+ - \alpha)$ is the population analogue of what the estimator has to handle at finite sample size.

The mechanical problem. The naive estimator has a denominator $M - \lfloor M\alpha \rfloor$, which is an integer. The true tail probability is $M(1 - \alpha)$, which need not be. When $M\alpha$ is not integer (which is almost always, at least at the confidence levels relevant for risk management), these two numbers disagree, and the estimator targets a slightly different confidence level than the one requested.

A worked example makes the size of the effect visible. Take $M = 1,003$ simulations and a requested confidence level $\alpha = 0.995$. The true tail probability is $M(1 - \alpha) = 5.015$. The naive estimator drops the first $\lfloor M\alpha \rfloor = 997$ losses and averages the remaining six, so its denominator is 6 and the effective tail probability implied is $6/1003 \approx 0.006$. The estimator is averaging a slightly larger tail than the one nominally asked for, and since observations deeper in the tail are larger, including the extra observation pulls the average *down* toward the bulk of the distribution. The estimator is systematically biased downward; it underestimates ES at the nominal confidence level, and the underestimation grows as $\alpha \rightarrow 1$ because the tail itself gets thinner and each rounding of $M\alpha$ represents a larger relative mismatch.

The Acerbi–Tasche correction. Acerbi and Tasche (2002) observed that the fix is to stop rounding and to discretise the integral form of ES directly. Let $w = \lceil M\alpha \rceil$, i.e. round the quantile index *up* rather than down. The Acerbi–Tasche estimator is

$$\widehat{\text{ES}}_\alpha^{\text{AT}} = \frac{1}{M(1 - \alpha)} \left[(w - M\alpha) L_{(w)} + \sum_{i=w+1}^M L_{(i)} \right]. \quad (15)$$

The difference from the naive form comes down to three things. The denominator is the exact tail probability $M(1 - \alpha)$, not a rounded integer. The observation at the empirical quantile, $L_{(w)}$, enters with a fractional weight $w - M\alpha \in [0, 1)$ rather than being fully included or fully excluded. And the remaining observations $L_{(w+1)}, \dots, L_{(M)}$ enter with full weight. The effect is to split the atom at the quantile boundary exactly in the proportions that the integral definition of ES requires.

Continuing the example: at $\alpha = 0.995$ and $M = 1,003$, the ceiling is $w = 998$, the fractional

weight is $w - M\alpha = 0.015$, and the denominator is $M(1 - \alpha) = 5.015$. The estimator gives observation 998 a weight of 0.015, observations 999 through 1003 a weight of 1 each, and divides the sum by exactly 5.015. Numerator and denominator now describe the same tail: the 0.5% that the question asked for, no more and no less.

Numerical illustration. Table 4 compares the two estimators on the two-bond portfolio of Example 3.2, using $M = 1,003$ Monte Carlo draws (chosen deliberately so that $M\alpha$ is non-integer at every confidence level on the grid) and averaging over $R = 20,000$ independent replications. True ES values are computed analytically from the integral form.

Table 4. Naive (13) versus Acerbi–Tasche (15) ES estimators on the two-bond portfolio of Example 3.2 ($M = 1,003$ MC draws, $R = 20,000$ replications). The column “Frac.” is the fractional weight $w - M\alpha$, the contribution that the naive estimator ignores by rounding the denominator.

α	True ES	Naive mean	Naive bias	AT mean	AT bias	Frac.
0.950	103.2000	103.1264	−0.0736	103.1798	−0.0202	0.150
0.975	106.4000	106.1352	−0.2648	106.3615	−0.0385	0.075
0.990	116.0000	114.5014	−1.4986	115.9038	−0.0962	0.030
0.995	132.0000	126.5575	−5.4425	131.6598	−0.3402	0.015

The naive estimator shows the pattern predicted by the mechanical argument above: systematic negative bias that grows as α rises. At $\alpha = 0.995$ the bias is -5.44 , about 4% of the true ES. This is not sampling noise: averaging over 20,000 independent replications removes random error essentially completely, so the number reported is the bias itself. The Acerbi–Tasche estimator tracks the true value within sampling noise across the entire grid, with biases an order of magnitude smaller at every confidence level. The fractional column “Frac.” shows the weight that the naive estimator discards: at $\alpha = 0.995$ it ignores 0.015 units of probability mass, which is precisely the correction that Acerbi–Tasche reinserts.

Implications for backtesting and aggregation. First, in contexts where ES backtesting procedures do apply — the daily frequencies and short horizons of market-risk validation, and the higher-frequency / lower-confidence model-validation exercises that risk teams run on credit sub-portfolios (see Section 6) — a backtest applied to naive estimates on atomic distributions produces biased test statistics even when the underlying model is correctly specified. Second, when aggregating sub-portfolio ES figures into a total, the naive estimator can generate spurious violations of subadditivity at finite M , which is exactly the defect that ES was introduced to eliminate at the population level. This second issue is universal: it applies at any frequency, any horizon, and any confidence level, because it is a property of the estimator, not of the statistical context in which it is used. Both problems disappear under Acerbi–Tasche, which reproduces the integral form exactly at any finite sample size.

For the continuous ASRF portfolio used for validation in this paper, the two estimators are numerically indistinguishable and the naive form is innocuous. For applied work with real portfolios, where the natural and standard practice is loan-by-loan Bernoulli simulation, the loss distribution is atomic by construction, and the Acerbi–Tasche estimator is the one to use. The same holds for any analysis in which the large- N approximation degrades: low-default portfolios, small credit-insurance books, stressed scenarios with concentrated exposures, or segments with heterogeneous LGD and EAD that do not average out into a smooth tail. The correction is a

single line of code relative to the usual implementation.

5.5 The decreasing ES/VaR ratio

The counter-intuitive finding is that the ES/VaR ratio decreases with α : the gap between the tail average and the tail threshold narrows as the confidence level rises. At $\alpha = 0.95$ the ratio is 1.30; at $\alpha = 0.999$ it has fallen to 1.13 and continues to decline. In the limit $\alpha \rightarrow 1$ the ratio approaches unity.

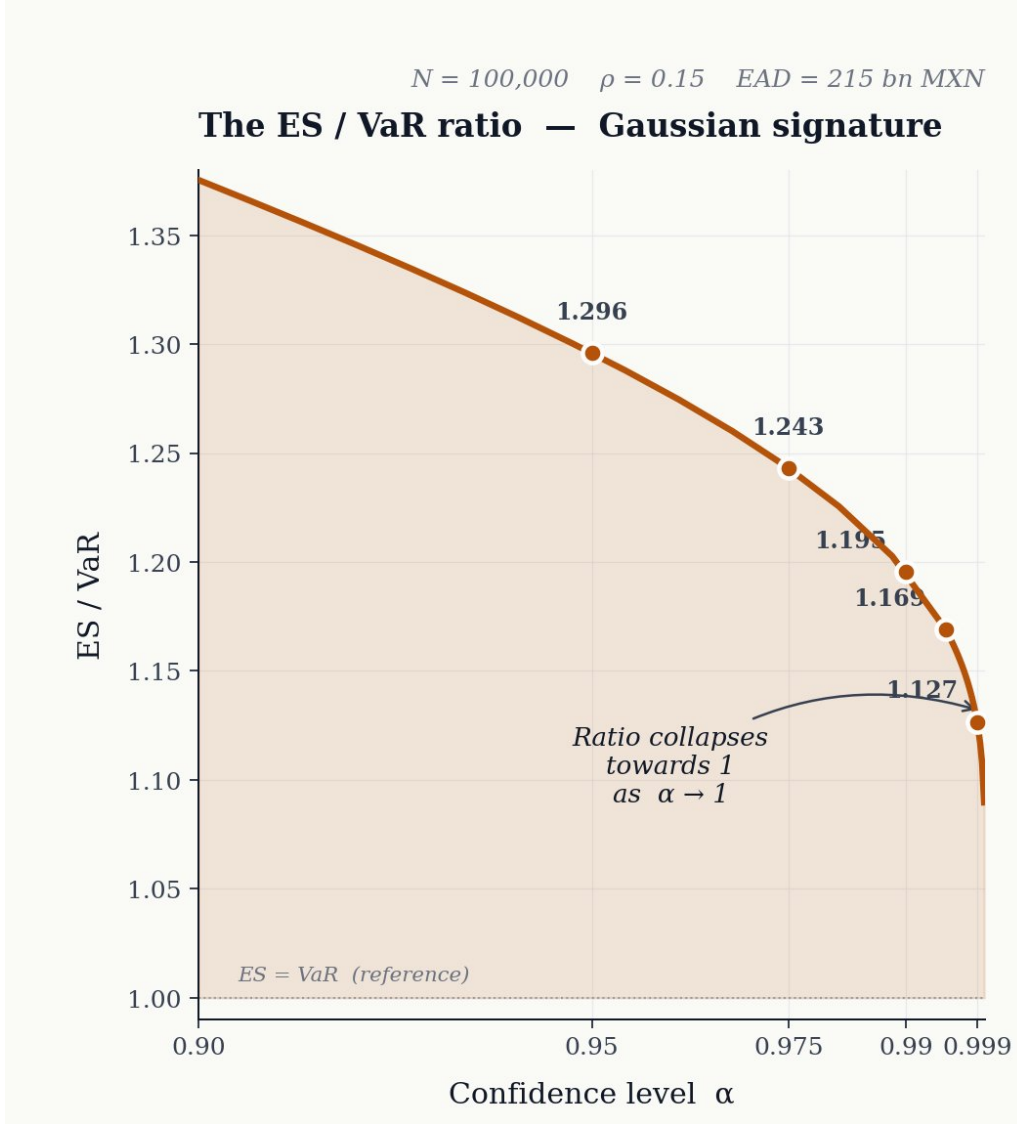


Figure 3. The ES/VaR ratio across confidence levels α for the 100,000-loan synthetic portfolio. The ratio declines monotonically toward unity as $\alpha \rightarrow 1$, a feature of the Gaussian factor assumption that does not survive richer dependence structures (Section 8). Marked points correspond to the values in Table 2.

The mechanism is the interaction of two functions. The conditional loss function $\mu_N(z)$ saturates as $z \rightarrow -\infty$: every borrower's conditional default probability approaches one and the portfolio loss approaches its maximum $\sum_i \ell_i$. The Gaussian density $\phi(z)$ decays super-polynomially in $|z|$. Deep in the tail the integrand $\mu_N(z)\phi(z)$ is dominated by the density factor, so the tail average collapses toward the tail threshold. Figure 4 makes the geometry visible: VaR is a single evaluation of μ_N at z_α ; ES is the density-weighted average of the same function over the entire

shaded tail region $z < z_\alpha$. As $\alpha \rightarrow 1$ the threshold z_α moves into the region where μ_N has nearly saturated, so the average over the tail and the value at the threshold become indistinguishable.

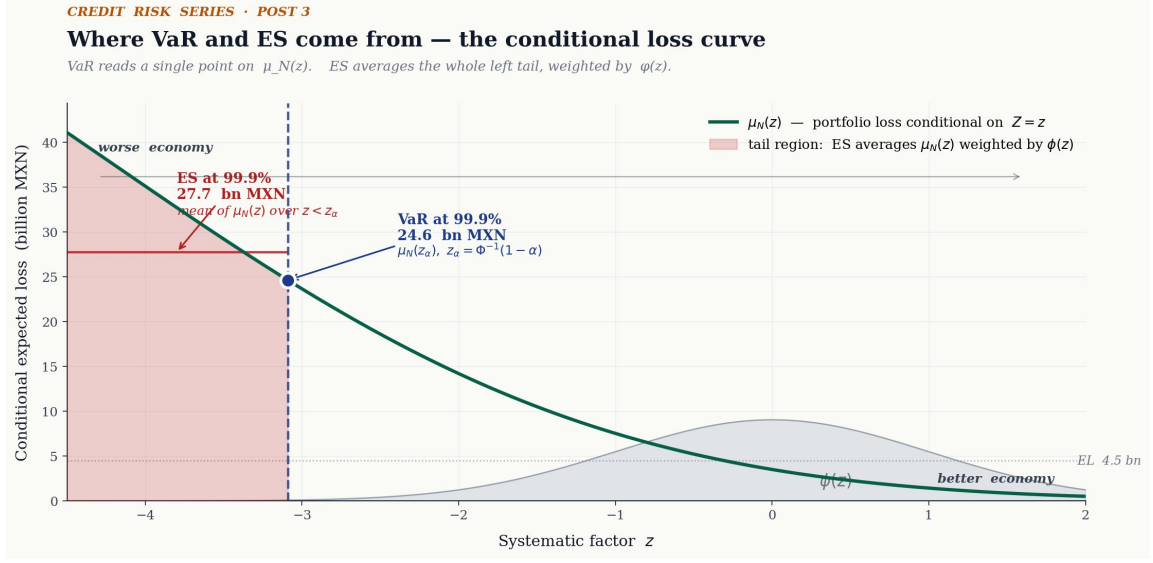


Figure 4. The conditional loss function $\mu_N(z)$ for the 100,000-loan portfolio. The systematic factor z is the common macroeconomic shock in the ASRF model: it has a standard normal distribution, and its effect on default is such that *negative* realisations of z raise every borrower’s conditional default probability. Large negative values of z therefore represent adverse macroeconomic states (“worse economy”, left panel), and large positive values represent favourable ones (“better economy”, right panel). VaR at 99.9% is the point value $\mu_N(z_\alpha)$ with $z_\alpha = \Phi^{-1}(0.001) \approx -3.09$, i.e. three standard deviations into the adverse tail; ES is the density-weighted average of $\mu_N(z)$ over the shaded region $z < z_\alpha$. The visual makes plain why the ES/VaR ratio collapses toward unity at high confidence levels: the curve μ_N saturates in the deep tail, so the average over the tail region differs less and less from the threshold value.

The mathematical statement is

$$\lim_{\alpha \rightarrow 1^-} \frac{\text{ES}_\alpha(L)}{\text{VaR}_\alpha(L)} = 1, \quad (16)$$

which holds under any factor model in which μ_N saturates and the systematic factor has a thin-tailed distribution.

Both conditions (saturation of μ_N in the deep tail and a thin-tailed factor distribution) are features of the Gaussian specification. They are not universal. Section 8 returns to this point to delimit which of the results in this paper are genuinely about credit portfolios and which are artefacts of a particular dependence assumption.

6 The Regulatory Migration from VaR to ES

6.1 How VaR became the standard, and why it was replaced

Banking regulation introduced VaR as the cornerstone of trading book capital with the 1996 Market Risk Amendment. For over a decade the approach was widely accepted as state-of-the-art: VaR was intuitive, computationally tractable, and consistent with the internal risk-management practices of large dealer banks. The limitations documented in the academic literature, notably the failure of subadditivity (Artzner et al., 1999) and the sensitivity of VaR estimators to distributional assumptions (Embrechts et al., 2002), were recognised but did not generate regulatory response.

The 2007–2009 financial crisis exposed practical failures of sufficient magnitude to force recon-

sideration. Losses at major dealer banks repeatedly exceeded one-day VaR estimates, in some cases by an order of magnitude, and the methodology’s insensitivity to tail shape meant that stressed scenarios produced only modest changes in reported VaR while the underlying tail expectations deteriorated dramatically. The Basel Committee responded with the Fundamental Review of the Trading Book consultative document in 2012 (Basel Committee on Banking Supervision, 2013), which proposed replacing VaR with Expected Shortfall at 97.5% as the primary market risk measure. The final standard was adopted in January 2019 (Basel Committee on Banking Supervision, 2019) with implementation deadlines that have now passed in all major jurisdictions.

6.2 Why 97.5%? The equivalence argument

The choice of 97.5% for ES in FRTB was not arbitrary. Under the assumption of a Gaussian loss distribution, a direct calculation shows that ES at the 97.5% level closely matches VaR at the 99% level. If $L \sim \mathcal{N}(0, \sigma^2)$, then

$$\text{VaR}_\alpha(L) = \sigma \Phi^{-1}(\alpha), \quad \text{ES}_\alpha(L) = \sigma \frac{\phi(\Phi^{-1}(\alpha))}{1 - \alpha}. \quad (17)$$

Numerical evaluation yields

$$\text{VaR}_{0.99}(L) = 2.326 \sigma, \quad \text{ES}_{0.975}(L) = 2.338 \sigma,$$

so that under Gaussian dynamics the two measures coincide to three decimal places. The Basel Committee set 97.5% precisely to achieve approximate calibration neutrality: under the benchmark Gaussian model, the transition from VaR 99% to ES 97.5% would leave capital requirements unchanged (Basel Committee on Banking Supervision, 2013). The upside would arise under non-Gaussian tails: when the true loss distribution has fatter tails than the Gaussian, ES 97.5% exceeds VaR 99% in a predictable direction, capturing risk that the quantile measure misses.

The same equivalence relationship applied to the ASRF portfolio of Section 5 gives a useful reality check. The portfolio ES at 97.5% is 17.24 bn MXN while VaR at 99% is 16.99 bn MXN, a gap of about 1.5%, close to the Gaussian approximation but not identical because the loss distribution of an ASRF portfolio is not exactly Gaussian. The gap widens under heavy-tailed factor models, which is exactly the direction FRTB intended to capture.

6.3 Backtesting and the elicibility question

A practical concern with the adoption of ES was the absence of a direct backtesting procedure. VaR admits a simple binomial test: count exceedances, compare to the expected frequency. ES does not admit such a test in isolation, because it is not elicitable in the sense of Gneiting (2011).

The Basel Committee’s solution under FRTB is a two-step monitoring regime. VaR at 97.5% continues to be backtested under the same binomial framework that applied previously. In parallel, the expected shortfall is monitored through a ratio-comparison procedure at the desk level: if the ratio of the actual to the model-implied ES departs materially from unity across sufficient observations, a capital add-on is triggered. This procedure delivers the regulatory benefits of ES — sensitivity to tail shape, coherence — while preserving the ability to identify models that are systematically misspecified.

The market-risk setting is, however, the easy case. The measurement horizon is one business day,

exceedences are counted daily, and a typical bank accumulates the 250-plus observations needed for the Kupiec test within a single trading year (Kupiec, 1995). When the underlying portfolio is a credit book and the horizon is one year, as in the Basel II IRB framework, Solvency II, and the insurance frameworks discussed below, standard backtesting tools collapse. The reason is mechanical. A VaR model at the 99.9% one-year level expects one exceedence per thousand years. The Kupiec proportion-of-failures test at that level has essentially no power without hundreds of years of independent observations, and Christoffersen’s conditional-coverage test (Christoffersen, 1998) compounds the problem by requiring enough exceedences to test for clustering. Credit institutions never have that sample. Even a well-run retail portfolio with twenty years of history provides twenty one-year realisations, and typically at most one or two will have contained a material tail event.

The literature has responded along several lines. Frerichs and Löffler (2003) and a series of follow-up papers proposed evaluating credit portfolio VaR models by comparing realised distributions of loss rates against their model-implied analogues using all observations, not only exceedences. Berkowitz (2001) advanced density-based tests that map the full forecast distribution onto the uniform distribution under correct specification, recovering information from every observed loss rather than from the binary exceedence indicator alone. A related line, associated with Frey, McNeil, and collaborators, uses rank-correlation and Spearman-type statistics across pooled borrowers or pooled time periods to extract degrees of freedom from the cross-section when the time series is too short (McNeil et al., 2015). For low-default portfolios specifically, the Basel Committee has acknowledged the issue in supervisory guidance on credit risk models, recommending benchmarking, sensitivity analysis, and stress-test-based validation in lieu of formal exceedence counting.

The honest summary is that *direct* backtesting of credit portfolio VaR at the 99.9% one-year level is not feasible with data that any single institution has on its own balance sheet. What is feasible, and is what most risk teams actually do, is to validate model inputs (PDs, LGDs, correlations) separately against their respective benchmarks, and to challenge the model at higher frequency and lower confidence levels (for instance, monthly default-rate projections at the 95% level on sub-portfolios with sufficient exposures). The aggregate one-year tail number itself is typically justified by auditability of inputs, model documentation, and supervisory review rather than by a statistical test of the tail itself.

6.4 Parallel trajectories in insurance regulation

Insurance regulation has followed a different path. The European Solvency II framework, operational since 2016, measures the Solvency Capital Requirement (SCR) as the VaR of the change in basic own funds at a 99.5% confidence level over a one-year horizon (European Insurance and Occupational Pensions Authority, 2014). The scope matters: “basic own funds” reflects the firm’s full balance sheet, so the SCR is an aggregate capital requirement that absorbs *both* asset-side risks (market, counterparty default on investments) *and* liability-side risks (underwriting, reserving, catastrophe). For a mortgage credit insurer the liability-side component is structurally analogous to the probable maximum loss (PML) of the insured portfolio, and it typically dominates the aggregate number. Within the EU framework the Mexican Circular Única de Seguros y Finanzas adopts the same structure, with a 99.5% Value-at-Risk computed over the stressed change in own funds.

The Swiss Solvency Test, operational since 2006, adopted Expected Shortfall from inception (Swiss Federal Office of Private Insurance, 2006). The SST target capital is computed as ES at the

99% level of the change in risk-bearing capital over a one-year horizon, again an aggregate measure spanning asset and liability risks. In terms of the Gaussian equivalence (17), ES 99% corresponds approximately to VaR 99.63%, slightly more conservative than Solvency II under benchmark assumptions, and substantially more conservative under fatter tails.

Canada’s Office of the Superintendent of Financial Institutions (OSFI) takes a third approach, and one directly relevant to the present paper. Its Mortgage Insurer Capital Adequacy Test (MICAT) is the only major international framework with an explicit risk-based capital methodology calibrated specifically to mortgage insurance, rather than inherited from a broader insurance solvency regime. The MICAT sets its supervisory target at a *conditional tail expectation* (CTE) of 99% over a one-year horizon (Office of the Superintendent of Financial Institutions Canada, 2025). CTE is the actuarial term of art for the tail average above a given threshold, $\mathbb{E}[L \mid L \geq \text{VaR}_\alpha(L)]$, and under continuous loss distributions it coincides with Expected Shortfall as defined in (2). The two terms are used interchangeably in what follows. In effect, Canada applies at the mortgage-insurance level the same measure and confidence level the Swiss Solvency Test applies at the aggregate insurance level. This calibration choice is worth noting given the scarcity of international precedents for risk-based capital tailored specifically to mortgage insurance, and MICAT is effectively the template.

International comparison of regimes is summarised in Table 5. The divergence between market-risk convergence (now uniformly on ES) and insurance divergence (Solvency II on VaR, SST and MICAT on ES / CTE) is a notable feature of the current regulatory landscape.

Table 5. Risk measure and confidence level across selected regulatory regimes.

Regime	Domain	Measure	Level
Basel II IRB	Banking credit risk	VaR	99.9%
Basel III FRTB	Banking market risk	ES	97.5%
Solvency II (EU)	Insurance (aggregate)	VaR	99.5%
Swiss Solvency Test	Insurance (aggregate)	ES	99.0%
OSFI MICAT (Canada)	Mortgage insurance	CTE	99.0%

7 Applications in Credit Risk Measurement

The machinery developed in Sections 4–5 applies naturally to two questions that arise in the management of a mortgage insurance portfolio: the sizing of retention limits and the measurement of solvency capital. The illustrations below use the same synthetic portfolio of Section 5 and are intended to document how VaR and ES behave under the same book, not to prescribe regulatory practice.

7.1 Retention sizing

A mortgage insurer retains risk up to a maximum retention limit N per exposure and cedes the remainder to reinsurance. Varying N traces out a family of loss distributions for the retained book: small N implies low but frequent retained losses; large N implies fewer but more severe retained losses. The question facing the risk manager is how to select N so that the retained loss distribution remains consistent with available capital.

The quantile approach asks: at what level of N does VaR at some confidence level reach a prescribed fraction of capital? The expectation approach asks the same question with ES. The

two approaches can differ meaningfully, and the direction of the difference is informative.

Consider two retention levels $N_1 < N_2$. The loss distribution retained by the insurer under N_2 stochastically dominates that under N_1 : every loss threshold is at least as likely to be exceeded. VaR and ES both agree on this ordering. But the two measures can disagree on *how much* N_2 is riskier than N_1 , and the direction of the disagreement is informative.

The mechanism behind the disagreement is what might be called *plateau behaviour* in VaR. Recall that VaR at level α is a single quantile: it reads the loss distribution at one point, the point where the cumulative probability equals α . If the retained loss density happens to be low in the neighbourhood of that point (for instance, because N sits beyond a cluster of likely loss values but before the next cluster), then the quantile function is nearly flat there: small shifts in N move the whole distribution upward, but the quantile crosses a sparsely populated region and barely moves. VaR therefore reports almost the same number for retention levels that are in fact materially different in their tail exposure.

A concrete example makes the point. Suppose the retained loss distribution under N_1 has a smooth tail with most mass between 10 and 20 bn MXN, and the 99% quantile falls at 17 bn. If we raise retention to N_2 by reinsuring out a smaller slice of mass at the 30 bn level, the new retained distribution has more weight far in the tail (say a bump around 30–35 bn), but the 99% quantile is still at 17 bn because the bulk of the mass below that point has not changed. VaR reports no difference. ES averages over everything above the quantile, so it picks up the shifted mass at 30–35 bn and reports a materially higher number under N_2 . Whether this sensitivity is a feature or a nuisance depends on what the retention decision is meant to protect against: if the concern is catastrophic losses in the far tail, ES captures exactly what VaR obscures.

The policy consequence is that VaR-based retention sizing can produce a range of acceptable N values that look indistinguishable under the quantile metric but correspond to meaningfully different tail-expectation exposures. An ES-based procedure discriminates between them. For a mortgage credit insurer in particular, where tail events are associated with macroeconomic downturns rather than random borrower-level shocks, the shape of the retained tail is typically where the interesting information lives: exactly the region VaR is least equipped to resolve.

7.2 Solvency capital under three measures

Table 6 reports the risk measure implied by three regulatory regimes — Solvency II, the Swiss Solvency Test, and Basel III’s FRTB — applied to the same 100,000-loan portfolio.

Before reading the table, one caveat. The three regimes were designed for different risk types: Basel II IRB and FRTB sit on the banking side (credit-risk capital and market-risk capital respectively), while Solvency II and SST sit on the insurance side and measure aggregate firm-level solvency that spans asset-side risks, underwriting risks, and operational risks simultaneously. Applying the four measures to a single credit-loss distribution is not the comparison the regulators intended; it is a calibration exercise that asks what number each formula *would* produce if the portfolio were the only risk driver in scope. It is a useful comparison for building intuition about how VaR and ES behave at different confidence levels on the same underlying distribution, but it is not a statement about which regime is “right” for a given institution. A mortgage insurer is regulated under an insurance framework, not under FRTB, and its true capital charge reflects the full balance sheet rather than the retained mortgage portfolio alone.

Canada’s MICAT, discussed in Section 6 and included in Table 5, is the only framework among those compared that is calibrated specifically to mortgage insurance. Other jurisdictions, in-

cluding Mexico under the CUSF, subsume the mortgage-insurance business under the general insurance solvency apparatus rather than separating it out.

Table 6. Risk measures of the synthetic mortgage portfolio under three regulatory regimes. Losses in bn MXN. The measures are computed on the same credit-loss distribution for calibration purposes; see the note in the text.

Regime	Measure	Value (bn MXN)
Solvency II (EU)	VaR 99.5%	19.33
Swiss Solvency Test	ES 99.0%	20.31
Basel III FRTB	ES 97.5%	17.24

A few observations are worth drawing out.

First, the Swiss Solvency Test measure ($ES_{0.99} = 20.31$ bn MXN) exceeds the Solvency II measure ($VaR_{0.995} = 19.33$ bn MXN) by approximately 5%, despite operating at a *lower* nominal confidence level. A regulator concerned with adverse tail scenarios captures more risk under the Swiss regime at 99% ES than under the European regime at 99.5% VaR. This follows directly from the shape of the loss distribution: ES averages over the entire tail, while VaR identifies a single point within it.

Second, the FRTB measure ($ES_{0.975} = 17.24$ bn MXN) sits below both insurance measures, which is consistent with its calibration neutrality relative to VaR 99% rather than to higher insurance thresholds. Under a Gaussian-shaped tail, the 97.5% ES is approximately equal to the 99% VaR; the portfolio result (17.24 vs. 16.99 bn MXN) confirms the equivalence to within 1.5%. The calibration is designed so that FRTB does not raise total market-risk capital materially on average, while shifting the measure to one that is sensitive to tail shape.

Third, the ordering among the three measures depends on the shape of the tail of the loss distribution. Under the Gaussian factor model used here, the gap between VaR 99.5% and ES 99% is modest (roughly 5%) and signed in the direction predicted by the Gaussian equivalence in (17). This gap is a function of two things: the confidence levels, and the shape of the tail above the VaR threshold. The first is a regulatory choice; the second is not, and it can change substantially when the underlying dependence structure departs from the Gaussian assumption, which is the opening for Section 8 below.

7.3 A note on measurement coherence

The discussion above describes how three regulatory measures behave on a single portfolio; it does not recommend adopting one over another. Within any given jurisdiction the choice of risk measure is a function of legal framework, supervisory tradition, and industry practice, none of which is the province of a technical paper.

What a technical paper can observe is the direction of evolution. Market-risk regulation converged, over two decades and the experience of one systemic crisis, on Expected Shortfall. The drivers of that convergence (coherence under portfolio aggregation, sensitivity to tail shape, calibration neutrality at chosen confidence levels) are properties of the loss distribution rather than properties of the market-risk domain per se. They apply equally to credit and insurance portfolios, where heavy tails and portfolio aggregation are also relevant considerations. In insurance, the Swiss Solvency Test has operated on an ES framework since 2006, providing over fifteen years of supervisory experience with an ES-based capital regime.

Whether the arguments that moved market-risk regulation will, in time, inform insurance frameworks elsewhere is beyond the scope of this paper. The technical case is on the table and has been stable in the academic literature for over two decades.

8 A Caveat: The Gaussian Factor Assumption

All numerical results above rest on one structural assumption: the ASRF model with a Gaussian factor Z and Gaussian idiosyncratic shocks ε_i . This is the dependence specification underlying Basel II’s IRB framework, and it also underlies Solvency II’s standard formula for credit risk. The specification has known limitations, and some of the numerical results of this paper turn out to depend on it rather than on more general properties of credit portfolios. This section makes that dependence explicit.

8.1 Two properties of the Gaussian specification

The Gaussian factor model has two properties that jointly drive much of the tail behaviour documented in earlier sections. The first is the way the systematic factor Z concentrates probability mass: the density $\phi(z)$ decays very rapidly as $|z|$ grows, so extreme realisations of the common driver become very unlikely very fast. The second is more subtle and concerns how borrowers behave jointly in the deep tail. Under a Gaussian dependence structure, any two asset values A_i and A_j with non-degenerate correlation are *asymptotically independent* when conditioned on extreme realisations: the conditional probability that both fall below increasingly extreme thresholds goes to zero, no matter how strongly correlated the variables appear in the bulk of the distribution. In plain terms: Gaussian dependence says that even strongly correlated borrowers stop defaulting together once the stress is severe enough.

Neither property is innocuous. The first compresses probability mass onto a narrow region of adverse states, making deep-tail scenarios artificially improbable. The second turns off the co-movement of defaults precisely when co-movement matters most: systemic events. Empirically, credit defaults during housing downturns or currency crises do not exhibit this decoupling (Embrechts et al., 2002): the conditional probability of joint default tends to stay materially above zero in the extreme tail, not to decay toward it.

8.2 Which results in this paper depend on that assumption

The coherence argument of Section 3, the two-bond counterexample of Example 3.2, and the closed-form derivations of Section 4 are robust to the choice of dependence structure: they rest on properties of the risk measures themselves, not on any particular factor model. The regulatory observations of Section 6 on FRTB, Solvency II, and SST calibration levels are similarly independent of the Gaussian assumption.

The result that the ES/VaR ratio *declines monotonically toward unity* as $\alpha \rightarrow 1$ on the synthetic portfolio, documented in Section 5.5, is not robust. It arises from the conjunction of two mechanisms, both specific to the Gaussian specification: saturation of the conditional loss function μ_N in the deep tail (where every borrower’s conditional default probability approaches one), and the fast decay of the Gaussian density that puts negligible weight beyond the threshold. Dependence specifications with heavier-tailed factors, or with co-movement of defaults that persists in the extreme tail, weaken one or both of these mechanisms. The direction of the effect is intuitive: if deep-tail scenarios carry more probability, or if borrowers co-default more strongly when stressed, the tail average will sit further above the tail threshold. Whether and at what confidence levels the ES/VaR ratio stabilises, reverses, or remains monotone depends on the

specification and its calibration, and is a quantitative question rather than a qualitative one.

8.3 Applications in credit-insurance capital measurement

For a mortgage credit insurer the choice of dependence specification is not an academic matter. Capital measures that look adequate under the Gaussian model may behave very differently under specifications that assign more probability to simultaneous extreme defaults, which is exactly the empirical pattern of housing-market downturns. The three regulatory measures compared in Section 7 were computed under the Gaussian ASRF; the magnitudes reported there should therefore be read as baseline values under a specific dependence assumption, not as invariant properties of the portfolio. The gap between ES and VaR measures (and especially its evolution across confidence levels) can shift materially when the assumption is relaxed, and that shift is where the economic content of the choice between the two measures becomes most visible.

The next post in this series takes up the question on the same synthetic portfolio used here, contrasting the Gaussian baseline against alternative dependence specifications to identify where the numerical conclusions remain stable and where they change.

9 Conclusion

Expected Shortfall is a mathematically coherent risk measure in the sense of Artzner et al. (1999); Value-at-Risk is not. The coherence argument is formally rigorous, admits a minimal two-bond counterexample, and is over two decades old in the academic literature. Under the Asymptotic Single Risk Factor model, both measures admit closed-form expressions: VaR is a single evaluation of the conditional loss function $\mu_N(z_\alpha)$, while ES is the density-weighted integral of the same function over the tail. Direct numerical integration via adaptive quadrature delivers ES to machine precision in a fraction of a second for portfolios of 10^5 loans, and the same result is recovered by Monte Carlo simulation of the systematic factor to within 1% using $M = 2 \times 10^4$ scenarios.

The ratio ES/VaR decreases with the confidence level under the Gaussian ASRF assumption, approaching unity as $\alpha \rightarrow 1$. This is a property of the Gaussian factor model rather than of credit portfolios. Whether, and how, the behaviour changes when the dependence structure changes is a quantitative question that this paper flags but does not resolve.

The regulatory migration from VaR to ES in market risk, completed by the Basel Committee in 2019 with Basel Committee on Banking Supervision (2019), was motivated by the properties documented in this paper: coherence under portfolio aggregation, sensitivity to tail shape, and the calibration neutrality achieved by choosing ES 97.5% as the regulatory standard. Insurance frameworks are split between the Solvency II standard (VaR 99.5%) and the Swiss Solvency Test standard (ES 99%), with the latter providing fifteen years of supervisory experience with ES-based solvency capital.

The paper is a technical exposition. Policy conclusions about which measure any particular framework should adopt lie beyond its scope; the mathematical and empirical case underlying the migration in market risk is, however, documented and reproducible.

References

Carlo Acerbi and Dirk Tasche. On the coherence of expected shortfall. *Journal of Banking & Finance*, 26(7):1487–1503, 2002.

- Philippe Artzner, Freddy Delbaen, Jean-Marc Eber, and David Heath. Coherent measures of risk. *Mathematical Finance*, 9(3):203–228, 1999.
- Basel Committee on Banking Supervision. Fundamental review of the trading book: A revised market risk framework. Consultative document, Bank for International Settlements, October 2013.
- Basel Committee on Banking Supervision. Minimum capital requirements for market risk. Technical Report d457, Bank for International Settlements, January 2019.
- Jeremy Berkowitz. Testing density forecasts, with applications to risk management. *Journal of Business & Economic Statistics*, 19(4):465–474, 2001.
- Peter F. Christoffersen. Evaluating interval forecasts. *International Economic Review*, 39(4):841–862, 1998.
- Paul Embrechts, Alexander McNeil, and Daniel Straumann. Correlation and dependence in risk management: Properties and pitfalls. pages 176–223, 2002.
- Susanne Emmer, Marie Kratz, and Dirk Tasche. What is the best risk measure in practice? a comparison of standard measures. *Journal of Risk*, 18(2):31–60, 2015.
- European Insurance and Occupational Pensions Authority. The underlying assumptions in the standard formula for the solvency capital requirement calculation. Technical report, EIOPA, July 2014.
- Hergen Frerichs and Gunter Löffler. Evaluating credit risk models: A critique and a new proposal. Working paper, University of Frankfurt, Center for Financial Studies, 2003.
- Tilmann Gneiting. Making and evaluating point forecasts. *Journal of the American Statistical Association*, 106(494):746–762, 2011.
- Michael B. Gordy. A risk-factor model foundation for ratings-based bank capital rules. *Journal of Financial Intermediation*, 12(3):199–232, 2003.
- Paul H. Kupiec. Techniques for verifying the accuracy of risk measurement models. *Journal of Derivatives*, 3(2):73–84, 1995.
- Alexander J. McNeil, Rüdiger Frey, and Paul Embrechts. *Quantitative Risk Management: Concepts, Techniques and Tools*. Princeton University Press, Princeton, NJ, revised edition, 2015.
- Office of the Superintendent of Financial Institutions Canada. Mortgage insurer capital adequacy test (micat) guideline. Technical report, OSFI, 2025. Effective January 1, 2025.
- Swiss Federal Office of Private Insurance. Technical document on the swiss solvency test. Technical report, Federal Office of Private Insurance (FOPI), 2006.
- Oldrich Vasicek. The distribution of loan portfolio value. *Risk*, 15(12):160–162, 2002.